Yang Li（郦洋）

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Education

Shanghai Jiao Tong University (SJTU), Shanghai, China — Sep. 2022 – Present

Doctor (Upgraded from Master's Program), Computer Science and Engineering

Research Interests: (Mathematical) Combinatorial Optimization. Supervisor: Weinan E and Junchi Yan

- GPA: 3.83/4.0. Rank Reference: 3 out of 211 achieving the Graduate National Scholarship
- Courses: 90.0% at A level

Shanghai Jiao Tong University (SJTU), Shanghai, China — Sep. 2018 – Jul. 2022

Bachelor, School of Computer Science

- GPA: 91.03/100 (or 3.93/4.3), Rank: 3/129
- Foundation Courses: 73.33% above A, 40.00% above A+
- Subject Courses: 80% above A, 50.00% above A+

Publications

First-authored (in Chronological Order):

1. **Generation as Search Operator for Test-Time Scaling of Diffusion-based Combinatorial Optimization**
   Yang Li, Lvda Chen, Haonan Wang, Runzhong Wang, Junchi Yan
   Advances in Neural Information Processing Systems, NeurIPS 2025
2. **Generative Modeling Reinvents Supervised Learning: Label Repurposing by Predictive Consistency Learning**
   Yang Li, Jiale Ma, Yebin Yang, Qitian Wu, Hongyuan Zha, Junchi Yan
   Forty-Second International Conference on Machine Learning ICML 2025
3. **Unify ML4TSP: Drawing Methodological Principles for TSP and Beyond from Streamlined Design Space of Learning and Search**
   Yang Li, Jiale Ma, Wenzheng Pan, Runzhong Wang, Haoyu Geng, Nianzu Yang, Junchi Yan
   International Conference on Learning Representations, ICLR 2025
4. **Fast T2T: Optimization Consistency Speeds Up Diffusion-Based Training-to-Testing Solving for Combinatorial Optimization**
   Yang Li, Jinpei Guo, Runzhong Wang, Hongyuan Zha, Junchi Yan
   Advances in Neural Information Processing Systems, NeurIPS 2024
5. **MixSATGEN: Learning Graph Mixing for SAT Instance Generation**
   Xinyan Chen*, Yang Li*, Runzhong Wang, Junchi Yan (*: equal contribution)
   International Conference on Learning Representations, ICLR 2024
6. **T2T: From Distribution Learning in Training to Gradient Search in Testing for Combinatorial Optimization**
   Yang Li, Jinpei Guo, Runzhong Wang, Junchi Yan
   Advances in Neural Information Processing Systems, NeurIPS 2023
7. **IID-GAN: an IID Sampling Perspective for Regularizing Mode Collapse**
   Yang Li*, Liangliang Shi*, Junchi Yan (*: equal contribution)
   International Joint Conference on Artificial Intelligence, IJCAI 2023
8. **Improving Generative Adversarial Networks via Adversarial Learning in Latent Space**
   Yang Li, Yichuan Mo, Liangliang Shi, Junchi Yan
   Advances in Neural Information Processing Systems, NeurIPS 2022 (Spotlight Top 5%)

Other (in Chronological Order):

1. **StruDiCO: Structured Denoising Diffusion with Gradient-free Inference-stage Boosting for Memory and Time Efficient Combinatorial Optimization**

Yu Wang, Yang Li, Junchi Yan, Yi Chang
Advances in Neural Information Processing Systems, NeurIPS 2025

2. **ML4CO-Bench-101: Benchmark Machine Learning for Classic Combinatorial Problems on Graphs**
Jiale Ma, Wenzheng Pan, Yang Li, Junchi Yan
Advances in Neural Information Processing Systems, NeurIPS 2025
3. **Bridging Crypto with ML-based Solvers: the SAT Formulation and Benchmarks**
Xinhao Zheng, Xinhao Song, Bolin Qiu, Yang Li, Zhongteng Gui, Junchi Yan
Advances in Neural Information Processing Systems, NeurIPS 2025
4. **COExpander: Adaptive Solution Expansion for Combinatorial Optimization**
Jiale Ma, Wenzheng Pan, Yang Li, Junchi Yan
Forty-Second International Conference on Machine Learning ICML 2025
5. **UniCO: On Unified Combinatorial Optimization via Problem Reduction to Matrix-Encoded General TSP**
Hao Xiong, Wenzheng Pan, Jiale Ma, Wentao Zhao, Yang Li, Junchi Yan
International Conference on Learning Representations, ICLR 2025
6. **Learning Plaintext-Ciphertext Cryptographic Problems via ANF-based SAT Instance Representation**
Xinhao Zheng, Yang Li, Cunxin Fan, Huaijin Wu, Xinhao Song, Junchi Yan
Advances in Neural Information Processing Systems, NeurIPS 2024
7. **Benchmarking pto and pno methods in the predictive combinatorial optimization regime**
Haoyu Geng, Han Ruan, Runzhong Wang, Yang Li, Yang Wang, Lei Chen, Junchi Yan
Advances in Neural Information Processing Systems, NeurIPS 2024
8. **正线性约束组合优化问题的非自回归学习求解 | Learning to Solve Combinatorial Optimization under Positive Linear Constraints via Non-Autoregressive Neural Networks**
汪润中, 郦洋, 严骏驰, 杨小康 | Runzhong Wang, Yang Li, Junchi Yan, Xiaokang Yang
中国科学：信息科学 | SCIENTIA SINICA Informationis 2024.
9. **ACM-MILP: Adaptive Constraint Modification via Grouping and Selection for Hardness-Preserving MILP Instance Generation**
Ziao Guo, Yang Li, Chang Liu, Wenli Ouyang, Junchi Yan
International Conference on Machine Learning, ICML 2024 (Spotlight Top 3.5%)
10. **Molecule Generation for Drug Design: a Graph Learning Perspective**
Nianzu Yang, Huaijin Wu, Kaipeng Zeng, Yang Li, Junchi Yan
Fundamental Research, 2024

Experience

ReThinkLab, Shanghai Jiao Tong University, Shanghai, China — Jul. 2021 – Present

Researcher Supervisor: Prof. Junchi Yan, Prof. Weinan E

See Publications for research outputs.

- Researcher, having published 5+ first-authored papers in top-tier academic conferences to date.
- My research interests lie in neural combinatorial optimization.

Honors and Awards

NeurIPS 2024 Top Reviewer	2024
National Scholarship (top 0.2% in the nation)	2019
Graduate National Scholarship (top 1% in the CS department)	2023
Outstanding Graduate of Shanghai (top 3%)	2022
Huawei Fellowship (top 3%)	2020
HyperGryph Fellowship (top 3%)	2021
1st-Class Academic Excellence Scholarship (top 1%)	2019
Merit Student of Shanghai Jiao Tong University	2019, 2020, 2025
1st-Class Academic Scholarship for Graduate Students	2022

Miscellaneous

- Travel History after the COVID-19 Pandemic: Hong Kong (2023), Macau (2023), United States (2023), South Korea (2024), Japan (2024), Hong Kong (2024), Singapore (2025)
- Academic Service: I serve as the reviewer for top-tier conferences, e.g., NeurIPS, ICML, ICLR, and journals, e.g., TPAMI.
- GitHub: https://github.com/yangco-le
- Languages: English - Experienced, Mandarin - Native speaker